Summary of Major Trade Models

(Ricardian, Specific-Factors, Heckscher–Ohlin, and Standard Trade Models)

Overview

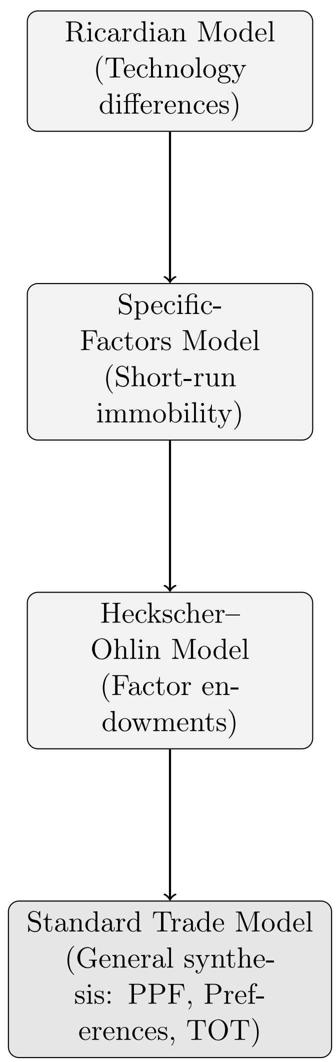
International trade theory evolves through four core models. Each adds complexity by modifying assumptions about **factors of production**, **mobility**, and **sources of trade**.

Model	Source of Trade	Factors of Production	Factor Mobility	Key Insights
Ricardian	Technological differences	Labour only	Labour mobile across sectors	Comparative advantage; all workers gain.
Specific-Factors	Short-run immobility of capital/land	Labour, capital, land (sector-specific)	Labour mobile; others fixed	Trade benefits export-sector owners, hurts import-sector owners.
Heckscher–Ohlin	Relative factor endowments	Labour and capital	Both mobile	Countries export goods using abundant factors; factor price equalization.
Standard Trade Model	Combination of technology, endowments, and preferences	Generalized factors	Flexible mobility assumption	Links production, demand, and terms of trade; unifies previous models.

Income Distribution Effects

- **Ricardian:** No inequality; all workers gain.
- **Specific-Factors:** Export-sector owners gain, import-sector owners lose.
- **Heckscher–Ohlin:** Abundant factor gains, scarce factor loses (Stolper–Samuelson).
- **Standard Trade Model:** Distributional effects depend on bias of growth and movement in terms of trade.

Conceptual Evolution Diagram



The Standard Trade Model synthesizes all prior frameworks by linking production (PPF), demand (indifference curves), and world prices (terms of trade).

Mathematical Foundations of the Major Trade Models

1. Ricardian Model

Assumptions:

- Labour is the only factor; technology differs across countries.
- Constant unit labour requirements a_{LX}, a_{LY}.
- Perfect competition $\Rightarrow$ goods prices reflect labour costs.

Equations:

$$\begin{aligned} L_X + L_Y &= \bar{L}, \\ a_{LX}X + a_{LY}Y &= \bar{L}, \\ \frac{P_X}{P_Y} &= \frac{a_{LX}}{a_{LY}} \quad \text{(in autarky)}. \end{aligned}$$

Trade occurs when relative prices differ:

$$\frac{a_{LX}}{a_{LY}} \neq \frac{a_{LX}^*}{a_{LY}^*}.$$

Result: Specialization by comparative advantage; world production and welfare rise.

2. Specific-Factors Model

Assumptions:

- Two goods: M (manufactures) and A (agriculture).
- Labour L is mobile; capital K specific to M, land T specific to A.

Production functions:

$$Q_M = Q_M(K, L_M), \quad Q_A = Q_A(T, L_A), \quad L_M + L_A = \bar{L}.$$

Wage equalization:

$$P_M\, MPL_M = P_A\, MPL_A = w.$$

Implications:

- An increase in P_M/P_A shifts labour toward M.
- r_K (return to capital) rises; r_T (return to land) falls.
- Labour's real wage effect is ambiguous.

3. Heckscher–Ohlin Model

Assumptions:

- Two factors: L and K; both mobile.
- Two goods: X (labour-intensive) and Y (capital-intensive).
- Identical technologies across countries; different factor endowments.

Zero-profit conditions:

$$P_X = a_{LX}\, w + a_{KX}\, r,$$
$$P_Y = a_{LY}\, w + a_{KY}\, r.$$

Given P_X/P_Y, solve for w/r.
Key Theorems:

- **Heckscher–Ohlin:** Countries export goods using their abundant factor intensively.
- **Stolper–Samuelson:** A rise in the price of a good raises the real return to the factor used intensively in that good.
- **Rybczynski:** An increase in one factor endowment increases output of the good using that factor intensively.

4. Standard Trade Model

Purpose: Combine production, demand, and world prices in one unified framework.
Relative supply (RS) and relative demand (RD):

$$RS = \frac{Q_X}{Q_Y} = f\left(\frac{P_X}{P_Y}\right), \qquad RD = \frac{D_X}{D_Y} = g\left(\frac{P_X}{P_Y}\right).$$

Equilibrium:

$$RS = RD \quad \Rightarrow \quad \frac{P_X}{P_Y} = \left(\frac{P_X}{P_Y}\right)^W.$$

Terms of Trade (TOT):

$$\text{TOT} = \frac{P_{\text{exports}}}{P_{\text{imports}}}.$$

Welfare: Improvement in TOT $\Rightarrow$ outward pivot of trade line $\Rightarrow$ higher indifference curve. Growth or policy shifts RS/RD, altering equilibrium prices and welfare.
Graphically: Welfare at tangency of highest indifference curve with world price line tangent to PPF.